SMART VEHICLE TRACKING AND ROUTE MONITORING SYSTEM USING IoT AND MOBILE APPLICATION

A MINI PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRICAL AND ELECTRONICS ENGINEERING

NPR COLLEGE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai) Natham,

Dindigul – 624 401

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NPR COLLEGE OF ENGINEERING AND TECHNOLOGY
(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this mini project report "**SMART VEHICLE TRACKING AND ROUTE MONITORING SYSTEM USING IoT AND MOBILE APPLICATION**" is the Bonafide work of "MARUTHACHALA MOORTHY C (583223105014), ANWAR KHAN A (583223105003)" who carried out the project work under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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DECLARATION

We, **MARUTHACHALA MOORTHY C (583223105014), ANWAR KHAN A (583223105003),** here by declare that the Creative and Innovative Mini Project report entitled "**SMART VEHICLE TRACKING AND ROUTE MONITORING SYSTEM USING IoT AND MOBILE APPLICATION"** submitted by ourselves in partial fulfillment of the requirement for the award of the degree of bachelor of Engineering, from NPR College of Engineering and Technology (An Autonomous Institution) Affiliated to Anna University, Chennai), Natham, Dindigul is my/our own work. The report has not been submitted for the award of any other degree / diploma of this university or any other university before.

Place: NPRCET

Date:

MARUTHACHALA MOORTHY C ,(583223105014)

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NPR COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

INSTITUTE VISION & MISSION

Vision

To develop students with intellectual curiosity and technical expertise to meetthe global needs.

Mission

- To achieve academic excellence by offering quality technical education using best teaching techniques.
- To improve Industry – Institute interactions and expose industrial atmosphere.
- To develop interpersonal skills along with value-based education in a dynamic learning environment.
- To explore solutions for real time problems in the society.

DEPARTMENT VISION & MISSION

Vision

- To equip students with technical skills and industry knowledge in Electrical and Electronics Engineering to meet global challenges.

Mission

- To provide an academic environment with strong theoretical and practical engineering skills
- To establish robust partnerships with industries for training, internships, and skill development.
- To nurture professionalism ethics,and acommitmentto societal well-being through value-based education.
- To develop globally competent Electrical and Electronics Engineers who would investigate, design, and offer solutions to society using cutting-edge technologies

PROGRAM OUTCOMES(POs)

Engineering Graduates will be able to:

PO 1: Engineering Knowledge: Apply the Knowledge of Mathematics, Science, Engineering Fundamentals, to the Solution of Complex EngineeringProblems
PO 2: ProblemAnalysis: Identify, formulate, review research literature, and analyze complex electrical engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineeringsciences
PO 3: Design/Development of Solutions: Design and develop electrical and electronic systems that meet specified needs with appropriate consideration for public health and safety, cultural, societal and environmental considerations.
PO 4: Conduct Investigations of complex problems: Investigate and analyze complex engineering problems using Research-based knowledge and research methods including design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
PO 5: Modern Tool usage: Select and Apply modern engineering and IT tools for simulation and modeling of electrical engineering systems**.**
PO 6:The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities to the professional engineering practice.
PO 7: Environment and Sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PO G: Individual and Team work: Function actively and efficiently as an individual or a member/leader of different teams and multidisciplinary projects.
PO 10 :Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.
PO 11: Project Management and finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12: Life-long Learning: Recognize the need for and have the preparation and ability to engage in independent and life- long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

Graduates of Electrical and Electronics Engineering Program will be able

PEO1: Have successful technical and professional careers in their chosen fields such as circuit theory, Field theory, control theory and computational platforms.

PEO2: Engross in life long process of learning to keep themselves abreast of new developments in the field of Electronics and their applications in power engineering.

PEO3: The graduates of the program will have capability in provided that best possible solutions to public problems with a comprehensive attitude.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Apply the modern tools and provide solutions for the real time problems related to energy conservation.

PSO2: Apply, develop and design the electrical and electronics systems used in power control applications and renewable energy industries with ethical standards.

23EE611 - Mini Project

Course Outcomes:		
On completion of the course, the students will be able to		Bloom's Level Mapped
CO1	On Completion of the mini project work students will be in a position to take up their final year project work and find solution by formulating proper methodology	Evaluate K5

(Action verb of each CO to be matched with the next mapping table)

Cos/Pos /PSOs MAPPING

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	POG	PO10	PO11	PSO1	PSO2
CO1	3	3	3	3	3	3	3	3	3	3	3	3	3
CO2	3	3	3	3	3	3	3	3	3	3	3	3	3
CO3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO4	3	3	3	3	3	3	3	3	3	3	3	3	3
CO5	3	3	3	3	3	3	3	3	3	3	3	3	3

(Mapping: 1- Low, 2-Medium, 3-High)

ABSTRACT

The rapid advancement of Internet of Things (IoT) technologies has significantly transformed modern transportation systems by enabling real-time monitoring, improved safety, and efficient vehicle management. Vehicle tracking systems have become an essential component in applications such as fleet management, logistics, public transportation, and personal vehicle security. However, most conventional tracking systems are limited to providing only real-time location information and lack intelligent functionalities such as route monitoring, trip counting, and alert generation. These limitations reduce their effectiveness in ensuring operational efficiency and safety. To address these challenges, this project proposes a smart vehicle tracking and route monitoring system that integrates IoT technologies with both web and mobile application interfaces to provide a comprehensive and user-friendly solution.

The proposed system is developed using an ESP32 microcontroller integrated with a GPS module to collect real-time latitude and longitude data of the vehicle. The collected data is transmitted to a cloud platform such as ThingSpeak using Wi-Fi communication, while GSM communication is used as a backup to ensure continuous data transmission in areas with limited internet connectivity. A web-based application is designed using HTML, CSS, and JavaScript along with Google Maps API to visualize the live vehicle location and route status. Additionally, a mobile application is developed using MIT App Inventor to provide a portable and accessible interface for users to monitor the system. Both applications retrieve data from the cloud and display real-time information including vehicle position, trip count, and system status.

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CHAPTER 1: INTRODUCTION

1.1 Overview

The rapid advancement of technology has led to the development of intelligent systems that enhance efficiency, safety, and automation across various sectors. One of the most significant areas of improvement is the transportation sector, where vehicle tracking systems play a crucial role in monitoring and managing vehicle movement. With the increasing number of vehicles and the demand for efficient transportation systems, there is a growing need for smart solutions that provide real-time tracking along with additional features such as route monitoring and trip analysis.

Vehicle tracking systems are widely used in applications such as fleet management, logistics, public transportation, and personal vehicle security. These systems help in reducing operational costs, improving safety, and enhancing overall efficiency. However, traditional systems often lack advanced functionalities required for intelligent monitoring.

1.2 Background of the Study

Traditional vehicle tracking systems primarily rely on Global Positioning System (GPS) technology to determine the location of a vehicle. While GPS provides accurate positioning information, many existing systems are limited to displaying location data without providing deeper insights into vehicle behavior. Most systems do not support features such as route detection, trip counting, or real-time alert mechanisms.

In addition, many conventional systems depend on a single communication method such as GSM or Wi-Fi, which limits their performance in areas with poor connectivity. These limitations make it difficult to use such systems in real-world applications where reliability and continuous monitoring are essential.

1.3 Problem Identification

Despite the availability of vehicle tracking systems, several challenges still exist in current implementations. One of the major issues is the lack of intelligent route monitoring. Most systems cannot determine whether a vehicle is following a predefined route or deviating from it. This reduces the effectiveness of monitoring in applications such as public transportation and fleet management.

Another major challenge is GPS inaccuracy. Due to environmental factors such as signal obstruction and atmospheric conditions, GPS data may contain small errors. Direct comparison of coordinates may lead to incorrect detection of locations, resulting in inaccurate trip counting.

1.4 Objectives of the Project

The main objective of this project is to design and develop a smart vehicle tracking and route monitoring system using IoT technologies. The specific objectives include:

- To implement real-time vehicle tracking using GPS
- To develop a route detection system based on start and end points
- To implement automatic trip counting
- To provide alert mechanisms for abnormal conditions
- To develop both web and mobile applications for monitoring
- To ensure reliable communication using Wi-Fi and GSM

1.5 Significance of the Project

The proposed system provides a comprehensive solution for vehicle tracking and monitoring. It enhances safety by providing real-time alerts and improves efficiency by automating trip counting and route detection. The use of both web and mobile applications increases accessibility and usability.

The system is cost-effective and easy to implement, making it suitable for various applications. It also demonstrates the potential of IoT technologies in developing smart transportation systems.

1.6 Organization of the Report

This report is organized into several chapters. Chapter 1 provides an introduction to the project. Chapter 2 presents the literature survey. Chapter 3 discusses system analysis. Chapter 4 describes system requirements. Chapter 5 explains the system implementation. Chapter 6 presents results and discussion. Chapter 7 concludes the project and suggests future enhancements.

CHAPTER 2
LITERATURE SURVEY

2.1 Introduction

The development of vehicle tracking systems has gained significant attention in recent years due to the increasing need for safety, efficiency, and real-time monitoring in transportation systems. With the rapid growth of the Internet of Things (IoT), various researchers have proposed solutions that integrate GPS, GSM, and cloud technologies to improve tracking capabilities. This chapter reviews existing systems and research works related to vehicle tracking, route monitoring, and IoT-based solutions.

2.2 GPS-Based Vehicle Tracking Systems

Global Positioning System (GPS) technology is widely used for vehicle tracking due to its ability to provide accurate location information. Several researchers have developed GPS-based tracking systems that use satellites to determine the position of a vehicle in real time..

2.3 GSM-Based Communication Systems

GSM (Global System for Mobile Communication) is commonly used in vehicle tracking systems for transmitting data over long distances. Many systems use GSM modules such as SIM800L to send location data via SMS or GPRS to a remote server. This allows users to monitor vehicle location even in areas where internet connectivity is limited.

2.4 IoT-Based Vehicle Tracking Systems

The introduction of IoT has significantly improved the capabilities of vehicle tracking systems. IoT-based systems use microcontrollers such as Arduino or ESP32 to collect data from sensors and transmit it to cloud platforms for

processing and visualization. Platforms such as ThingSpeak and Firebase are commonly used to store and analyze data in real time.

2.5 Route Monitoring and Trip Analysis Systems

Some advanced vehicle tracking systems incorporate route monitoring and trip analysis features. These systems use predefined routes and compare them with real-time GPS data to determine whether a vehicle is following the correct path..

2.6 Mobile and Web-Based Monitoring Applications

Modern vehicle tracking systems often include mobile and web applications for user interaction. Mobile applications provide portability and allow users to monitor vehicle location from anywhere, while web applications offer better visualization through maps and dashboards.

MIT App Inventor is widely used for developing mobile applications due to its simple drag-and-drop interface. Similarly, web technologies such as HTML, CSS, and JavaScript are used to create interactive dashboards with map integration using Google Maps API.

2.7 Need for the Proposed System

To overcome the limitations of existing systems, there is a need for an advanced vehicle tracking system that integrates real-time monitoring, route detection, and alert mechanisms. The proposed system aims to provide a cost-effective and scalable solution using IoT technologies.

CHAPTER 3
SYSTEM ANALYSIS

3.1 Introduction

System analysis is an important phase in the development of any project, as it helps to understand the existing system, identify its limitations, and propose an improved solution. In the context of vehicle tracking, traditional systems provide only basic location information and lack advanced functionalities such as route monitoring, trip counting, and alert mechanisms. This chapter discusses the analysis of the existing system, its drawbacks, and the proposed system that overcomes these limitations.

3.2 Existing System

The existing vehicle tracking systems are mainly based on GPS and GSM technologies. These systems use a GPS module to collect location data and transmit it to the user via GSM communication or web platforms. The user can view the vehicle's location using a mobile application or web interface.

Although these systems are widely used, they have several limitations. Most existing systems only provide real-time location tracking and do not include features such as route detection or trip counting.

3.3 Disadvantages of Existing System

The following are the major disadvantages of existing vehicle tracking systems:

- Limited to basic location tracking
- No route monitoring capability
- No automatic trip counting
- Dependence on a single communication method

- Inaccurate results due to GPS signal fluctuations
- Lack of real-time alert system

These limitations reduce the effectiveness and reliability of traditional tracking systems in real-world applications.

3.4 Proposed System

The proposed system is a smart vehicle tracking and route monitoring system using IoT technologies. It integrates GPS, ESP32 microcontroller, cloud platform, and both web and mobile applications to provide an advanced tracking solution.

The system collects real-time GPS data and transmits it to a cloud platform such as ThingSpeak using Wi-Fi. GSM communication is used as a backup to ensure continuous operation. A web application and a mobile application are used to display the vehicle's location, trip count, and route status.

3.5 Advantages of Proposed System

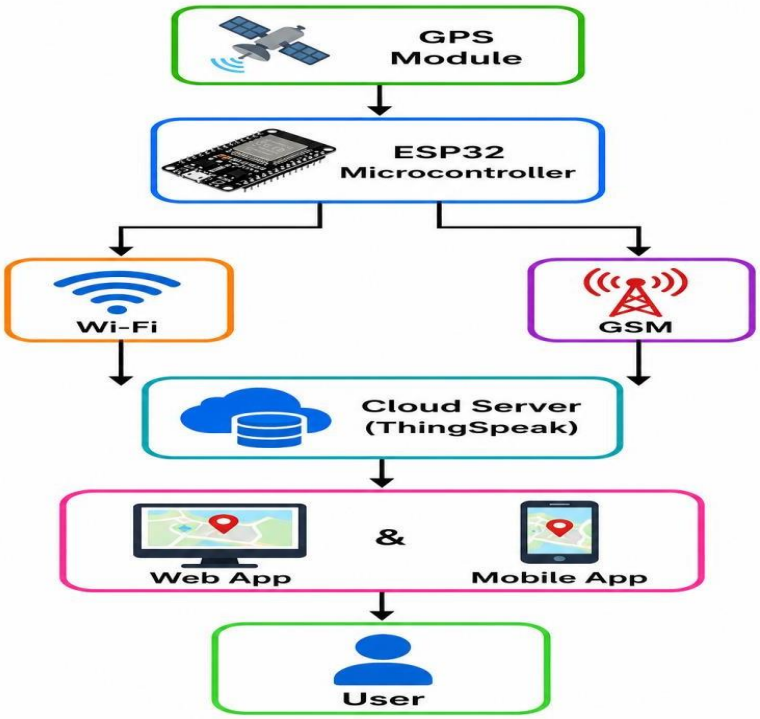
The proposed system offers several advantages:

- Real-time vehicle tracking
- Accurate route detection using tolerance method
- Automatic trip counting
- Dual communication (Wi-Fi + GSM)
- Web and mobile application support
- Alert system for abnormal conditions
- Cost-effective and scalable solution

3.6 System Architecture

The system architecture represents the overall structure and interaction between different components of the system.

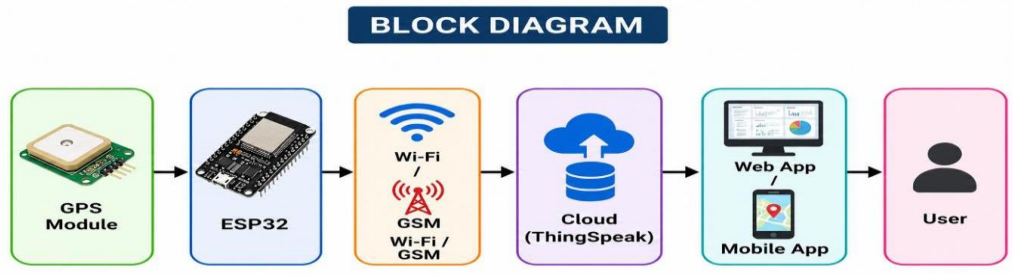
Architecture Diagram



3.1 Block Diagram

The block diagram represents the flow of data from input to output in the system.

Block Diagram



3.2 Working Principle

The working principle of the system is based on real-time data collection, transmission, and processing. The GPS module continuously collects location data in the form of latitude and longitude. This data is sent to the ESP32 microcontroller, which processes it and transmits it to the cloud platform.

The web and mobile applications fetch the data from the cloud and display it to the user. The system compares the current location with predefined start and end points using a tolerance-based approach. When the vehicle reaches the start location, the system begins tracking the trip. When the end location is reached, the trip count is incremented.

Alerts are generated if the vehicle deviates from the route or if there is a loss of GPS signal or communication.

CHAPTER 4
SYSTEM DESIGN

4.1 Introduction

System design is the process of defining the architecture, components, modules, and data flow of a system. It transforms the requirements identified in the system analysis phase into a structured solution. In this project, the system design focuses on integrating hardware components (GPS, ESP32), communication modules (Wi-Fi, GSM), cloud platform (ThingSpeak), and user interfaces (Web and Mobile App).

4.2 System Components

The proposed system consists of the following major components:

4.2.1 GPS Module

The GPS module is responsible for collecting real-time location data in the form of latitude and longitude.

4.2.2 ESP32 Microcontroller

ESP32 acts as the main controller. It processes GPS data and sends it to the cloud using Wi-Fi. It also handles GSM communication for backup.

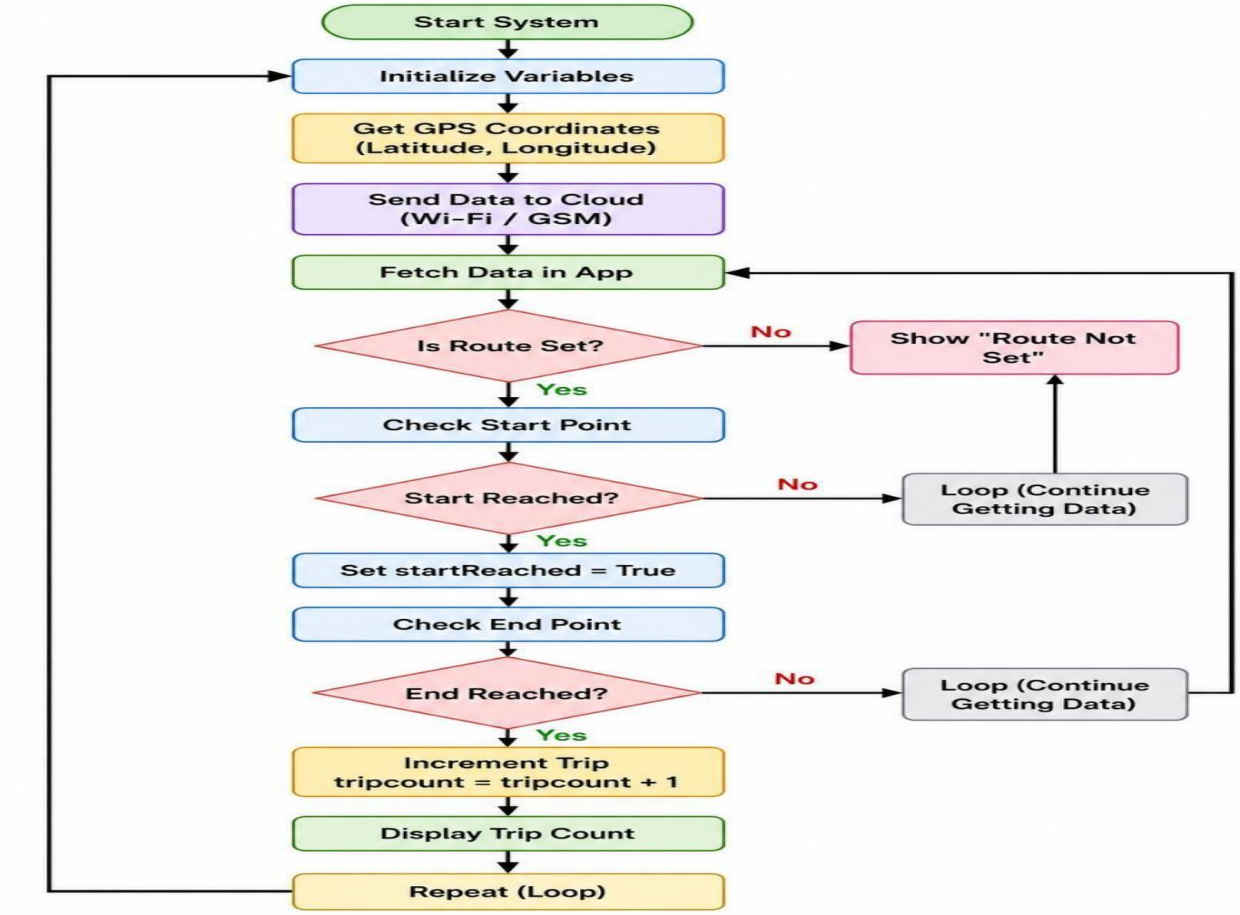
4.2.3 Communication Module

- **Wi-Fi:** Primary data transmission
- **GSM:** Backup communication

4.2.4 Cloud Platform (ThingSpeak)

Stores GPS data and allows retrieval through APIs.

4.3 Flowchart Diagram



CHAPTER 5: SYSTEM REQUIREMENTS

5.1 Introduction

System requirements define the essential specifications needed to develop and operate the proposed system effectively. These requirements include both hardware and software components, along with functional and non-functional requirements. A clear understanding of system requirements ensures that the system performs reliably and meets the intended objectives.

5.2 Functional Requirements

Functional requirements describe the core functionalities that the system must perform.

5.2.1 Real-Time Location Tracking

The system must continuously collect the vehicle's geographic location using the GPS module. The latitude and longitude values should be updated in real time and transmitted to the cloud platform.

5.2.2 Route Setting

The system must allow the user to input start and end coordinates manually through the mobile application. These coordinates are used as reference points for route monitoring.

5.2.3 Route Detection

The system should compare the real-time GPS coordinates with predefined start and end points using a tolerance-based approach. It must detect when the vehicle reaches the start and end locations.

5.2.4 Trip Counting

The system must automatically increment the trip count when the vehicle completes a journey from the start point to the end point.

5.2.5 Data Transmission

The system should transmit GPS data to the cloud platform using Wi-Fi. In case of Wi-Fi failure, GSM communication must act as a backup.

5.2.6 Data Visualization

The system must display real-time vehicle location on a map using a web application and mobile application.

5.2.7 Alert System

The system should generate alerts in the following situations:

- Route deviation
- GPS signal loss
- Communication failure

5.3 Hardware Requirements

The hardware components required for the system are listed below:

5.3.1 ESP32 Microcontroller

ESP32 is the core component of the system. It processes GPS data and handles communication. It is chosen due to its built-in Wi-Fi, low cost, and high performance.

5.3.2 GPS Module (NEO-6M)

The GPS module is used to collect real-time location data. It provides latitude and longitude information.

5.3.3 Connecting Wires and Accessories

Jumper wires and connectors are used for interfacing components.

5.4 Software Requirements

The software tools required for the system are as follows:

5.4.1 Arduino IDE

Used to program the ESP32 microcontroller.

5.4.2 ThingSpeak Cloud Platform

Used for storing and retrieving GPS data.

5.4.3 Web Technologies

- HTML (structure)
- CSS (design)
- JavaScript (logic)
- Google Maps API (map visualization)

5.4.4 MIT App Inventor

Used to develop the mobile application for tracking and monitoring.

5.4.5 Browser and Internet Connection

Required for accessing the web application and cloud platform.

CHAPTER 6
SYSTEM IMPLEMENTATION

6.1 Introduction

System implementation is the phase where the proposed design is transformed into a working model. It involves integrating hardware components, developing software programs, and establishing communication between different modules. In this project, the implementation focuses on building a real-time GPS vehicle tracking system using ESP32, integrating it with a cloud platform, and developing both web and mobile applications for monitoring and control.

6.2 Hardware Implementation

The hardware implementation involves interfacing the GPS module and GSM module with the ESP32 microcontroller.

6.2.1 ESP32 Programming

The ESP32 is programmed using Arduino IDE. The program reads GPS data, extracts latitude and longitude values, and sends them to the ThingSpeak cloud platform using HTTP requests

6.2.2 Mobile Application Development (MIT App Inventor)

The mobile application is developed using MIT App Inventor. It provides a simple interface for users to input start and end coordinates.

The application uses components such as:

- LocationSensor (for GPS data)
- Web component (to fetch data from ThingSpeak)
- Labels and Buttons (for display and control)

6.3 Route Detection Logic Implementation

The core functionality of the system is route detection and trip counting. The logic is implemented as follows:

- The user sets start and end coordinates
- The system continuously receives current GPS location
- If the current location is close to the start point (within tolerance), the system sets startReached = true
- If startReached = true and the current location matches the end point, the trip is considered complete
- The trip count is incremented and displayed

This tolerance-based approach ensures accurate detection even when GPS values fluctuate slightly.

6.4 Data Communication

Data communication is achieved using HTTP protocol. The ESP32 sends GPS data to ThingSpeak using API keys.

CHAPTER 7
RESULTS AND DISCUSSION

7.1 Results

The developed GPS-based vehicle tracking system was successfully implemented and tested under real-time conditions. The system effectively collected location data using the GPS module and transmitted it to the cloud platform using the ESP32 microcontroller.

The route detection logic performed effectively using the tolerance-based approach. When the vehicle reached the predefined start location, the system correctly identified it and updated the status to “Start Reached.”

CHAPTER 8
ADVANTAGES

The proposed GPS vehicle tracking system offers several advantages compared to traditional tracking methods. One of the primary advantages is real-time tracking capability. The system continuously updates the vehicle's location, allowing users to monitor movement at any time. This feature is highly useful for fleet management, logistics, and personal vehicle security.

CHAPTER 9
FUTURE SCOPE & CONCLUSION

The proposed system can be further enhanced by integrating advanced technologies. One possible improvement is the use of machine learning algorithms to predict vehicle movement and optimize routes. This would make the system more intelligent and efficient.

Another future enhancement is the integration of real-time alerts and notifications using mobile push notifications or SMS. This would help users receive immediate updates in case of route deviation or emergency situations.

In this project, a GPS-based vehicle tracking and route monitoring system was successfully designed and implemented using IoT technologies. The system integrates hardware components such as ESP32 and GPS modules with cloud platforms, web applications, and mobile applications to provide a complete tracking solution.

Overall, the project demonstrates a practical and cost-effective solution for modern vehicle tracking applications. It can be used in various domains such as transportation, logistics, and personal vehicle monitoring. The system meets the project objectives and provides a strong foundation for future enhancements.

Cos/Pos /PSOs MAPPING

COs/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	**2**	**1**	**1**	**2**	**1**	**1**	**1**	**1**	**2**	**1**	2	**2**	**1**
CO2	3	3	3	**2**	3	**1**	**1**	**1**	**2**	**2**	**2**	2	3	**3**
CO3	3	3	**2**	3	3	**1**	**1**	**1**	**2**	**2**	**2**	2	3	**3**
CO4	**2**	3	**2**	3	**2**	**2**	**1**	**2**	**2**	**2**	**2**	3	**2**	**2**
CO5	**2**	**2**	3	**2**	3	**1**	**1**	**1**	3	3	**2**	3	**2**	**3**

(Mapping: 1- Low, 2-Medium, 3-High)

COs–SDGMAPPING

COURSE OUTCOME	DESCRIPTION	MAPPED SDGS	JUSTIFICATION
C407.1	Conduct a literature survey to identify real-world problems and propose methodology for GPS vehicle tracking and monitoring system	SDG 4 – Quality Education SDG 9 – Industry, Innovation and Infrastructure	Enhances research skills, promotes technical knowledge in IoT and embedded systems, and supports innovative transportation monitoring solutions
C407.2	Design and develop the GPS vehicle tracking system using ESP32, GPS module, cloud platform, and web technologies	SDG 9 – Industry, Innovation and Infrastructure SDG 11 – Sustainable Cities and Communities	Encourages smart transportation systems and efficient vehicle monitoring using modern technologies
C407.3	Build and demonstrate the project effectively as a team	SDG 9 – Industry, Innovation and Infrastructure SDG 17 – Partnerships for the Goals	Promotes teamwork, collaboration, and practical implementation of IoT-based real-time tracking solutions
C407.4	Evaluate route deviation, tracking challenges, and implement alert mechanisms for safety	SDG 11 – Sustainable Cities and Communities SDG 12 – Responsible	Focuses on safer transportation, route optimization, and responsible monitoring systems

COURSE OUTCOME	DESCRIPTION	MAPPED SDGS	JUSTIFICATION
		Consumption and Production	
C407.5	Communicate project outcomes through presentations, reports, and demonstrations	SDG 4 – Quality Education SDG 17 – Partnerships for the Goals	Develops communication skills and supports effective sharing of technical knowledge and project results

COs vs WKS Mapping Table

COURSE OUTCOME	DESCRIPTION	MAPPED WKS ATTRIBUTES	JUSTIFICATION
C407.1	Conduct a literature survey to identify problems and propose methodology	WKS1 – Engineering Knowledge WKS4 – Investigation	Applies engineering fundamentals, literature review, and analytical methods for solving transportation monitoring problems
C407.2	Develop and implement the GPS tracking system using hardware and software tools	WKS2 – Problem Analysis WKS5 – Modern Tool Usage	Involves embedded systems, cloud platforms, and web technologies for real-time monitoring
C407.3	Build and demonstrate the project as a team	WKS6 – The Engineer and Society WKS9 – Individual and Team Work	Encourages collaboration and application of engineering solutions for public transportation safety
C407.4	Evaluate risks, route deviations, and corrective alert mechanisms	WKS2 – Problem Analysis WKS7 – Environment and Sustainability WKS11 – Project	Focuses on risk management, safety monitoring, and efficient project execution
C407.5	Present project results using reports and demonstrations	WKS10 – Communication	Enhances technical communication and documentation skills

POs / PSOs Mapping Justification

TYPE OF PROJECT	REAL TIME PROJECT
TITLE OF THE PROJECT	IoT-Based GPS Vehicle Tracking and Route Monitoring System

PO / PSO	JUSTIFICATION
PO1	The project applies engineering fundamentals such as GPS communication, embedded systems, IoT connectivity, and web technologies using ESP32 and cloud platforms.
PO2	The problem of inefficient vehicle monitoring and route tracking is identified and analyzed. Real-time tracking and route deviation detection are implemented to improve transportation monitoring.
PO3	A complete system consisting of ESP32, GPS module, ThingSpeak cloud, and web dashboard is designed for real-time tracking and alert generation.
PO4	The project involves real-time GPS data collection, route analysis, trip counting, and performance evaluation through live monitoring.
PO5	Modern engineering tools such as HTML, JavaScript, Leaflet.js, ThingSpeak, and ESP32 programming environments are used for implementation.
PO6	The system improves public and vehicle safety through live monitoring and route deviation alerts, benefiting transportation management.

PO / PSO	JUSTIFICATION
PO7	Efficient route monitoring helps reduce unnecessary fuel consumption and improves transportation efficiency, supporting sustainable practices.
PO8	Ethical engineering practices are followed in handling GPS data, cloud communication, and secure monitoring systems.
PO9	The project involves teamwork in hardware integration, coding, testing, and website development, demonstrating collaborative skills.
PO10	Technical concepts and project outcomes are communicated through reports, presentations, web dashboards, and demonstrations.
PO11	Cost-effective components such as ESP32 and open-source technologies are used, ensuring efficient project management and resource utilization.
PO12	The project demonstrates continuous learning through IoT, cloud integration, GPS systems, and web development technologies.
PSO1	The project applies concepts of embedded systems, communication protocols, and electronics engineering in developing a real-time vehicle monitoring system.
PSO2	Modern software and cloud tools such as ThingSpeak, Leaflet.js, JavaScript, and ESP32 programming are used to solve real-world transportation problems.

REFERENCES

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